TradeAlerts Simulation/Backtest Engine – 2025-08-06

What Is This Project?

This is a real-time, indicator-driven stock trading simulation and backtest platform. It pulls live quotes from the E*TRADE API, applies a configurable set of technical analysis indicators and risk management logic, and simulates the buying and selling of equities according to your custom strategy. It maintains all trade and position history in a local SQLite database and features a Flask dashboard for live monitoring, simulation, and review.

How Does It Work?

1. Universe Scanning

- Scans a universe of symbols (S&P500 by default) every N seconds.
- For each symbol:
  - Fetches the latest price and recent history from E*TRADE.
  - Calculates all enabled indicators and checks if entry/exit signals fire.
  - If buy signals fire and you have cash, a simulated buy is executed.
  - If risk or time-based exit conditions trigger, a sell is executed.

2. Trade Execution

- All trades are recorded to a `simulation_trades` table.
- Holdings are managed in a normalized `holdings` table.
- Cash is deducted/credited on buy/sell with full P/L tracking.

3. Dashboard

- Live status of all holdings (qty, price paid, last price, gain/loss).
- Trade history with time, symbol, action, qty, price, and P/L.
- All indicators and risk management settings are reflected live.

Available Technical Indicators

From your config (`simulation_config.json`) and code (`indicators.py`), these are currently implemented and available:

Trend/Price Action

- Simple Moving Average (SMA): Configurable length (default: 8 and 20).
- Price above/below SMA20: Optional “must be present” filter.
- Range breakout filter: Checks for % range breakouts.

Momentum

- Relative Strength Index (RSI): Length, overbought/oversold, and slope.
- MACD: Fast, slow, and signal, plus MACD histogram slope.

Volatility

- Bollinger Bands (BB): Length and stdev. Can require price breakout from band.
- ATR (Average True Range): With both absolute and percentage thresholds.
- Gap filter: Percent-based gap-up/gap-down entry logic.

Volume

- Volume spike: Compares volume to an N-day average.

VWAP

- VWAP filter: Checks price proximity to VWAP.

ADX (Average Directional Index)

- ADX length and threshold required for “trending” condition.

Supertrend

- Parameters: fast, slow, signal.

News filter

- Toggleable, currently disabled in your config.

Signal Combination

- Requires a configurable minimum number of signals (e.g. 3) for any buy.

Pyramiding

- Max pyramids per symbol (default: 3, for scaling in).

Risk Management (From `risk_management.py` & `simulation_config.json`)

Multiple layers of risk management are implemented:

Per-Trade Risk

- max_per_trade: $150 — No single buy can exceed this dollar amount.
- starting_cash: $1000 (for backtest/sim).

Position Management

- stop_loss_pct: 5% — Sells any position that falls more than 5% below cost.
- take_profit_pct: 12% — Sells for profit when a position gains 12%.
- trailing_stop_pct: 12% — If enabled, maintains a trailing stop based on high-water mark.

Max Hold Days

- sell_after_days: 1 — Sells any position held longer than this many days (hard exit).

Signal Strictness

- min_signals: 3 — Requires this many signals to buy.
- strict_buy_signals: 4 — For extra filtering.

Other Controls

- single_entry_only: Disabled, so re-entries allowed up to max pyramids.
- nuke_db: Disabled, so database is not auto-wiped.
- pause_when_market_closed: True, respects market hours.

Current Settings (From `simulation_config.json`)

Setting	Value
SMA	On, len=8
RSI	On, len=7, OB=65, OS=35
MACD	On, 6/19/4
BB	On, len=10, std=1.5
Volume Spike	On, multiplier=2.5
VWAP	On, threshold=1.0
ADX	On, len=14, threshold=20

Setting	Value
ATR	On, threshold=2.0, pct=0.5
Supertrend	On, 7/21/5
Trailing Stop	On, 12%
Stop Loss	On, 5%
Take Profit	On, 12%
Sell After Days	1
max_per_trade	150
starting_cash	1000
min_signals	3
strict_buy_signals	4
max_pyramids	3
require_sma20	true

- News filter: Off
- Backtesting, simulation, and alert logic all use these settings.

Risk Management Code Highlights

- Exit Logic: See `trading_helpers.py` (`check_exit_orders`), which will:
    - Sell on stop loss, take profit, or after N days.
    - Uses live E*TRADE quotes for exits.
- Trade Recording: Both buys and sells are recorded to `simulation_trades` table for full auditability.
- Holdings: Always up-to-date, P/L always recalculated with live prices.

Key Features / Workflow

1. All indicators and settings are controlled by `simulation_config.json` (drop-in editable).
2. Every trade is simulated as if live, using only available cash.
3. Automatic exit and risk logic means no “bag holding.”
4. Backtesting and real-time simulation use same logic and settings, for one-to-one comparison.
5. Dashboard fully reflects current positions, P/L, trade history, and settings.

What's Next / Areas to Check

- Confirm you're seeing sells and trade history after the next market open.
- Confirm that `trading_helpers.py` and the simulation dashboard are reflecting sold positions correctly (should update instantly after a sell event).
- You can fine-tune indicator strictness, risk management, or try new combos by just editing `simulation_config.json` and restarting the sim.

If you want a concise copy-paste version or a markdown doc, just say the word. Anything you want clarified or expanded, just let me know!